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Project Title:

AeroPassport: Flight Hardware Traceability and Lifecycle Management System

Technical Description:

Technology Stack

Backend Framework

- Java 21
- Spring Boot

Database

- MongoDB

API Documentation and Testing

- Springdoc OpenAPI
- Swagger UI
- Postman

Build and Dependency Management

- Apache Maven
- Maven Wrapper (mvnw)

IDE

- Antigravity IDE
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Database Design

The application consists of four collections.

1. Suppliers Collection

Stores supplier information.

Purpose:

- Maintain supplier records.
- Track origin of components.

2. Parts Collection

Stores individual flight hardware components.

Purpose:

- Maintain hardware inventory.
- Associate components with suppliers.
- Track operational status.

3. Part Passports Collection

Stores the digital identity of components.

Purpose:

- Maintain traceability.
- Store embedded inspection records.
- Store lifecycle events.
- Maintain readiness status.

4. Modules Collection

Stores subsystem assemblies.

Purpose:

- Group multiple components.
- Track module composition.

MongoDB Data Modeling Strategy

The project uses a hybrid document modeling approach.

→ Referencing

Examples:

Supplier → Parts

Part → Passport

Module ↔ Parts

Benefits:

- Reduced redundancy.
- Easier updates.

→ Embedding

Examples:

Passport

Inspections

Events

Benefits:

- Faster retrieval.
- Improved traceability.
- Atomic updates.

MongoDB Indexing Strategy

Indexes are implemented to improve query performance.

Collection	Indexed Field	Type
parts	supplierId	Single Field
parts	batchNumber	Single Field
part_passports	partId	Unique

modules	partIds	Multikey
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REST API Design

Base URL:

<http://localhost:8080/api>

→ Supplier APIs

Manage supplier records.

Method	Endpoint	Description
POST	/suppliers	Create supplier
GET	/suppliers	Retrieve all suppliers
GET	/suppliers/{id}	Retrieve supplier by ID
PUT	/suppliers/{id}	Update supplier
DELETE	/suppliers/{id}	Delete supplier

→ Part APIs

Manage hardware components.

Method	Endpoint	Description
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POST	/parts	Register new part
GET	/parts	Retrieve all parts
GET	/parts/{id}	Retrieve part
PUT	/parts/{id}	Update part
DELETE	/parts/{id}	Delete part

→ Module APIs

Manage subsystem modules.

Method	Endpoint	Description
POST	/modules	Create module
GET	/modules	Retrieve modules
GET	/modules/{id}	Retrieve module
PUT	/modules/{id}	Update module
DELETE	/modules/{id}	Delete module

→ Reporting APIs

Generate MongoDB-powered reports.

Hardware Genealogy Report - GET /api/parts/{partId}/genealogy

Description:

Returns the complete lifecycle and traceability history of a component, including:

- Part information
- Supplier details
- Passport details
- Embedded inspections
- Embedded events
- Current module assignment
- Flight readiness status
- Derived genealogy status

→ Administrative APIs

Database Statistics

GET /api/admin/statistics

Returns:

- Total suppliers
- Total parts
- Flight-ready parts
- Parts under inspection
- Total modules

→ Database Reset

DELETE /api/admin/reset

Purpose:

Drops application collections for testing and demonstration purposes.

Logging Strategy

Spring Boot logging is used to record major lifecycle actions.

Examples:

INFO Supplier created: SUP001

INFO Part registered: PAR001

INFO Passport generated: PAS001

INFO Event appended: Received

INFO Genealogy report generated.

Purpose:

- Console-based execution
 - Improve auditability.
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Verification and Testing Strategy

→ Swagger UI

Used for:

- Endpoint discovery.
- Interactive testing.

→ Postman

Used for:

- CRUD verification.
- Report validation.
- Administrative operation testing.

→ MongoDB Compass

Used for:

- Document inspection.
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